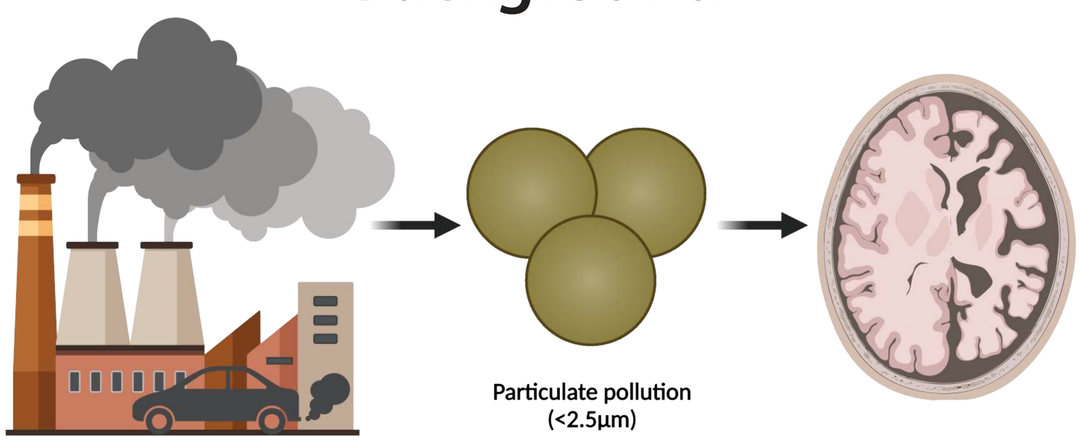
Predicting Change in County-Level Alzheimer's Disease Incidence Rate using Shapelet-Based Analysis of Pollution Recordings

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Background

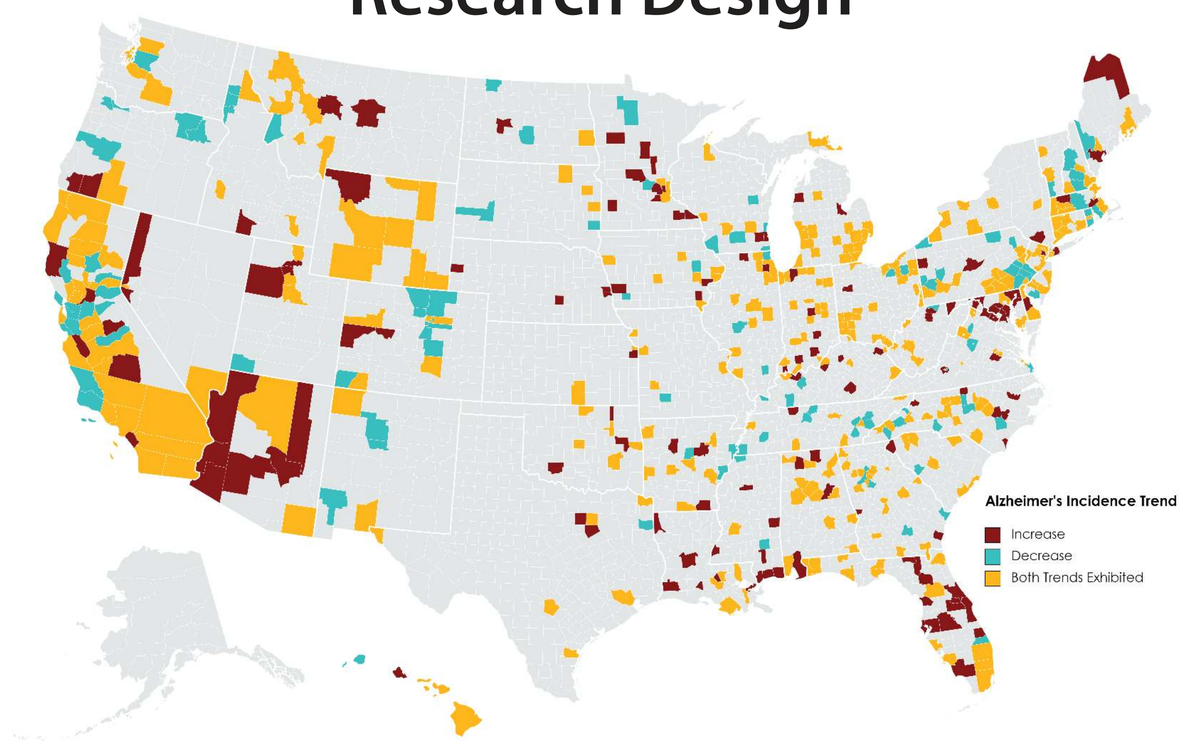


- One in three people over 65 die with Alzheimer's Disease (AD)[1].
- Increasing particulate pollution levels are a factor in the rise of AD [2,3].
- Particulate matter with a diameter of <2.5 µm (pm2.5) has been shown to have the greatest effect, but the influence of pm2.5 exposure patterns is yet to be investigated. A high average exposure to pm2.5 is harmful, but it is unclear whether there is a difference between the effects of variable and constant exposure.

Objectives

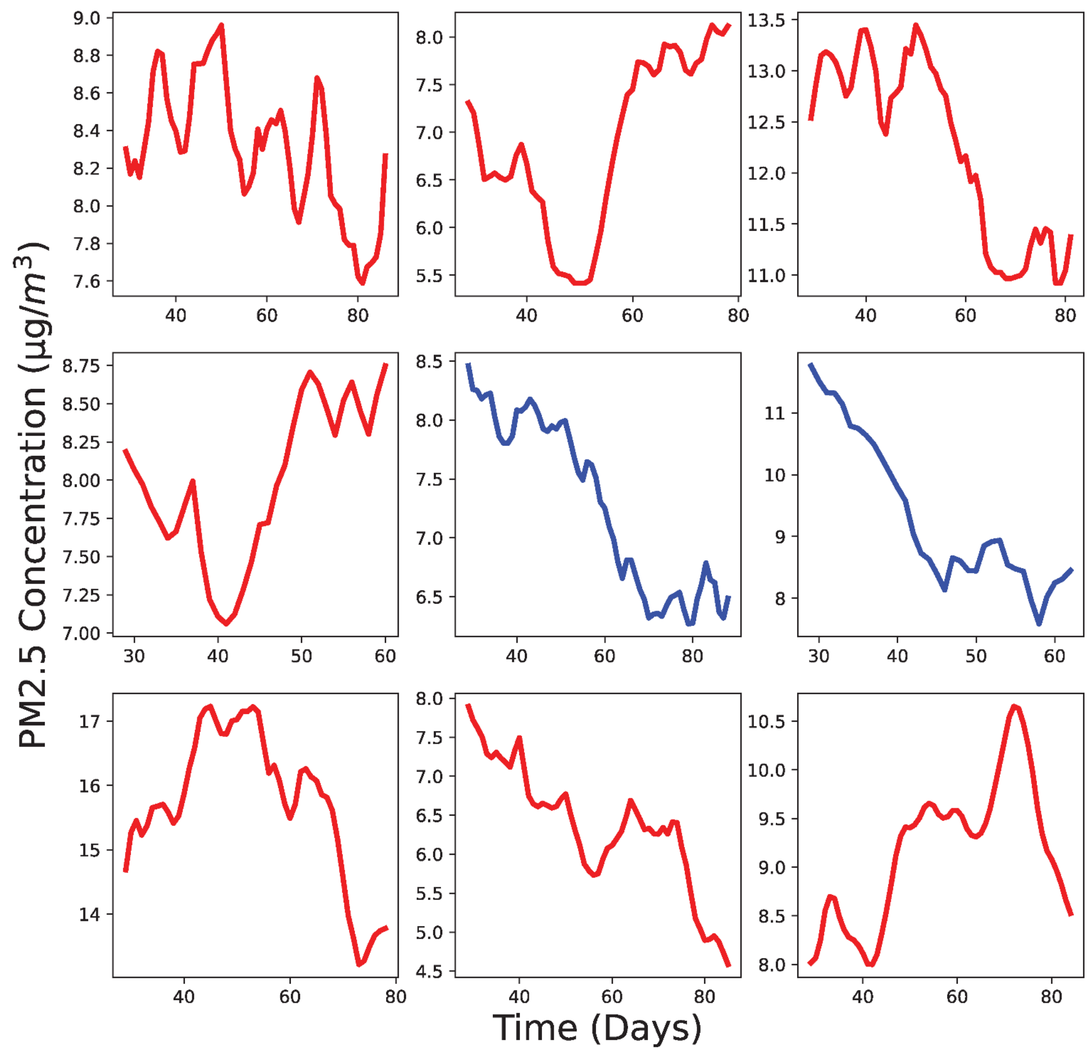
- **Determine if pm2.5 pollution patterns provide predictive power for Alzheimer's Disease incidence rate.**
- **Build an effective predictive model using data from the Centers for Medicare & Medicaid Services (CMS) and EPA.**
- **Extract most important patterns to help inform mechanism research and prevention methods.**

Research Design



- All publicly available daily average pm2.5 recordings from the Environmental Protection Agency's Air Quality System (AQS) from 2008-2017 were grouped by county to create a county daily average.
- Counties with over 9 days of consecutive missing data were eliminated.
- 2009-2018 County-level AD incidence rates for those >65 years were collected from the Centers for Medicare and Medicaid Services (CMS).
- Years where a county experienced an incidence rate in the top or bottom 25% were retained and labeled high/low.
- Each year of pollution data was used to predict the incidence rate trend for the following year.
- A shapelet transform classifier with shapelet length 5-90 days was used to create a predictive model and was validated using 5-fold cross validation.
- For comparison, yearly average pm2.5 concentration was calculated and used in a standard binary logistic regression to predict incidence rate trends.

Results



Top 9 Discriminatory Shapelets from PM2.5 Data– Shown in order of information gain, left-right and top-bottom. Red shapelets were associated with jumps in AD and blue with drops. Smoothed with a 30-day rolling window to eliminate noise and improve interpretability.

Binary Logistic Regression Model	Accuracy: 48%
Shapelet Transform Classifier	**Accuracy: 70%** **AUC: 0.76**

Conclusions

- The profile of pm2.5 levels over a year proved to be an effective predictor of AD incidence rate trends in counties in the U.S., especially given the complexity of AD development.
- Shapelet-based analyses are useful methods to understand temporal effects of exposures on various clinical conditions.
- High incidence rates appear to be associated with jumps in pm2.5 concentration, while low incidence rates seem to be associated with drops.
- If further investigation supports our results, local governments can focus their pollution and population health efforts towards reducing large jumps in pm2.5 concentration.

Future Plans

- Understanding the bidirectional effects of exposures and aging leveraging similar methods.
- Applying these techniques to other age-associated diseases such as Parkinson's Disease and COPD.
- Creating tools for healthcare systems to anticipate and prepare for population disease load.

References

1. Alzheimer's disease facts and figures [Internet]. Alzheimer's Disease and Dementia. Alzheimer's Association; [cited 2022Sep8]. Available from: https://www.alz.org/alzheimers-dementia/facts-figures
2. Power MC, Adar SD, Yanosky JD, Weuve J. Exposure to air pollution as a potential contributor to cognitive function, cognitive decline, brain imaging, and dementia: A systematic review of Epidemiologic Research. NeuroToxicology. 2016;56:235–53.
3. Shi L, Wu X, Danesh Yazdi M, Braun D, et al. Long-term effects of PM2·5 on neurological disorders in the American Medicare population: A longitudinal cohort study. The Lancet Planetary Health. 2020;4(12).

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